

TASK 3 - Project Requirement Documentation

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Chapter 1 - INTRODUCTION

The project **Wearable Seizure Detection and Alert System** is an system designed to help individuals with seizure problem. It continuously monitors the users **body movements and heart rate to detect abnormal patterns that may indicate seizures or falls**, and immediately alerts a caregiver to ensure timely assistance.

The main objective of this project is to provide a low-cost, real-time solution for **detecting tonic-clonic seizures, sudden falls, and abnormal heart rate events**, while also recording the data for future analysis. This wearable device ensures safety, faster response, and peace of mind for both patients and caregivers.

1.1 Purpose

The purpose of this project is to monitor seizure-related events and sudden falls in real-time, providing alerts to caregivers and storing event data for medical analysis. It aims to enhance safety, emergency response, and continuous monitoring for people .

1.2 Need / Motivation

Many people with epilepsy experience sudden seizures or falls that can cause injuries or delay medical assistance. Caregivers and family members **cannot constantly monitor patients, especially when alone**. Existing medical devices are often **expensive or hospital-based, making them less accessible**.

This wearable system solves these problems by providing a low-cost, continuous monitoring solution that detects seizure events, issues immediate alerts, and records important health data. It increases safety, independence, and confidence for patients while assisting caregivers in timely interventions.

Chapter 2 - OBJECTIVE

The objective of this assignment is to create a detailed specification document that outlines the functional requirements, non-functional requirements, and system architecture of the proposed engineering project. These elements form the foundation of the project design and help ensure that the system will meet both the user needs and technical constraints effectively.

In this assignment, we will be required to develop a Requirements Specification document that focuses specifically on three core aspects of your project:

Functional Requirements – What the system must do.

Non-Functional Requirements – How the system should perform.

System Architecture – The structure of the system and the design principles that will be followed.

The document should clearly define each of these areas and provide enough detail to guide the development of the system.

Chapter 3 - REQUIREMENTS

3.1 Functional Requirements (What the System Must Do)

Definition: Functional requirements describe the specific behaviour or functions of the system. These requirements define the core operations the system should be able to perform, typically from the perspective of end-users or stakeholders.

3.2 List of Core Functions: -

FR1: The system must continuously monitor body motion using an accelerometer and gyroscope sensor to detect abnormal movement patterns associated with **Tonic-Clonic Seizure**.

Priority: High

Acceptance Criteria: The system should sample motion data at a minimum of 50 Hz and detect repetitive shaking patterns within 3–5 seconds.

FR2: The system must detect sudden falls or abnormal orientation changes using motion sensor data.

Priority: High

Acceptance Criteria: The system should identify a fall event when sudden acceleration followed by orientation change is detected within 1–2 seconds.

FR3: The system must send an emergency alert to a caregiver or connected device when a seizure event is detected.

Priority: High

Acceptance Criteria: The alert notification should be transmitted within 5 seconds after seizure detection.

FR4: The system must activate a vibration alert on the wearable device to warn the user when abnormal motion patterns are detected.

Priority: Medium

Acceptance Criteria: The vibration motor should activate within 2 seconds after abnormal motion is detected.

FR5: The system must record sensor data during detected events in sleep also for further analysis and model training.

Priority: Medium

Acceptance Criteria: The system should store at least 10 seconds of motion data before and after a detected event.

ID	Functional Requirement	Priority	Acceptance Criteria
FR1	Detect Seizures	High	Identify abnormal rhythmic shaking
FR2	Fall Detection	High	Detect sudden fall impact.
FR3	Heart Monitoring	High	Measure abnormal heart rate changes before or during seizure.
FR3	Alert SOS	High	Notify caregiver via SMS or app notification.
FR4	Vibration alert	Medium	Notify the user if the seizer going to occur
FR5	Record Data	Medium	Store sensor readings for later analysis.

Figure 1 :- Table of Non-Functional Requirements

3.3 Non-Functional Requirements (How the System Should Perform)

Definition: Non-functional requirements define how the system should behave under various conditions and include aspects such as performance, security, usability, and maintainability. These requirements do not describe specific functions but instead focus on the **quality attributes** that the system must exhibit.

ID	Category	Requirement Description
NFR1	Safety	Use of low-voltage sensors to ensure user safety.
NFR2	Performance	Process sensor data and send alerts immediately.
NFR3	Reliability	Consistently detect seizure/fall events without false alarms.
NFR4	Usability	Comfortable wearable with simple, user-friendly interface.
NFR5	Maintainability	Easy to replace components or update firmware.
NFR6	Scalability	Allow integration of additional sensors in the future.

Figure 2 :- Table of Non-Functional Requirements

3.4 SYSTEM ARCHITECTURE (HOW THE SYSTEM IS STRUCTURED)

Definition: The system architecture defines the overall structure and design of the system, including its components, how they interact, and the technologies or frameworks used. It provides a high-level view of the system, focusing on key decisions made regarding its design and structure.

Instructions: Describe the system architecture that will guide the development of your project. This should include the high-level design, key components, and interactions.

Component	Description
Microcontroller	ESP32 processes sensor data, runs detection logic, triggers alerts, and stores events.
Motion Sensor	MPU6050 measures acceleration and rotation to detect seizures and falls via I2C.
Heart Sensor	MAX30102 monitors heart rate changes to detect pre-seizure signs via I2C.
Actuator	Buzzer or vibration motor alerts the user when abnormal activity is detected.
Communication	Bluetooth sends alerts to caregivers and mobile apps.
Power	Rechargeable battery powers all components for wearable use.

Table 3 :- Table of System Architecture

3.5 SYSTEM DIAGRAMS

System Block Diagram

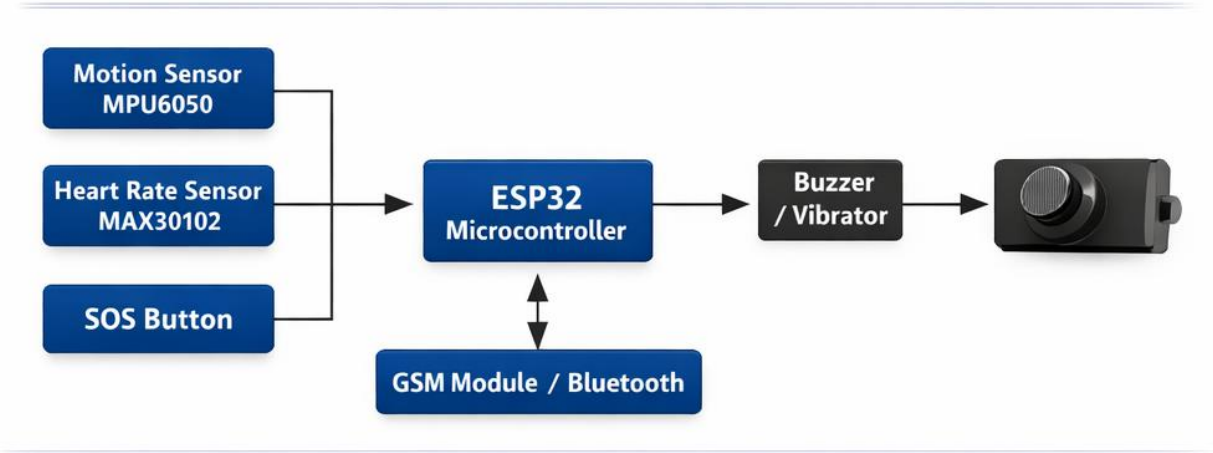


Figure 1 :- Block Diagram

Wiring Diagram

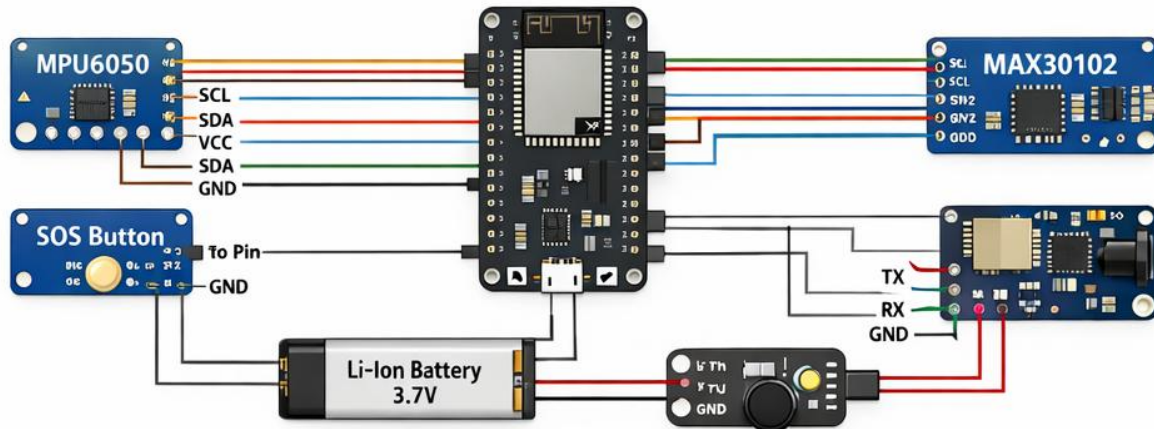


Figure 2: - Wiring Diagram

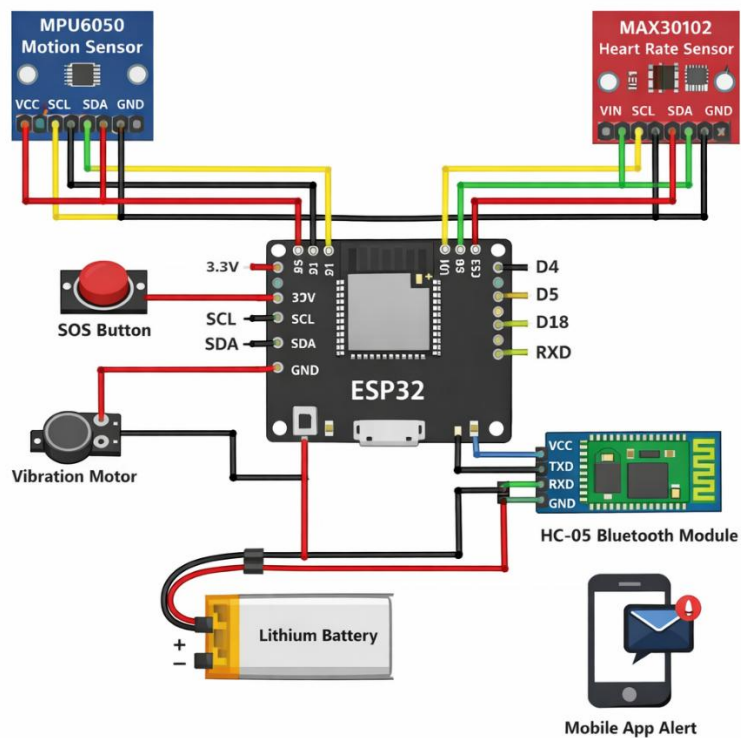


Figure 3: - Circuit Schematic.

3.6 HARDWARE REQUIREMENT

Microcontroller: ESP32
Motion Sensor: MPU6050
Heart Rate Sensor: MAX30102
Bluetooth/Wi-Fi module
Buzzer / Vibration motor
Push Button (SOS)
Rechargeable Battery
Optional: SD Card Module for data storage

3.7 SOFTWARE REQUIREMENT

Arduino IDE / PlatformIO for ESP32 programming
Libraries for sensors:
MPU6050 library
MAX30102 library
GSM/Bluetooth library
Mobile App (optional) for notifications
Data logging software (if storing to SD card or cloud)
Basic algorithm for seizure and fall detection

Chapter 4 - Conclusion

The Wearable Seizure Detection and Alert System is a practical, low-cost solution to improve the safety of people. By using sensors, the device can detect **tonic-clonic seizures**, sudden falls, and abnormal heart patterns, record the events, and immediately alert caregivers. It demonstrates how wearable technology can have a significant social impact, providing peace of mind for both patients and caregivers.

Future improvements could include advanced seizure prediction, additional sensor integration, and mobile app analytics to make the system even more effective.